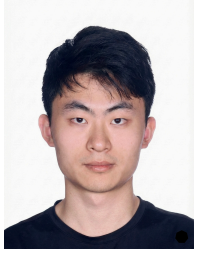
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Education and Employment

Beihang University Hangzhou Innovation Institute, Assistant Research Fellow (Postdoctoral Researcher) Jan. 2026 – Present

Postdoctoral advisors: Academician Jiancheng Fang and Prof. Jianli Li.

School of Instrumentation Science and Opto-Electronics Engineering, Beihang University, Ph.D. in Electronic Information Sep. 2020 – Nov. 2025

Supervisor: Prof. Zhuo Wang; research conducted with Academician Jiancheng Fang's team and the National Key Laboratory of Inertial Technology.

Shandong University of Science and Technology, B.Eng. in Measurement and Control Technology and Instrumentation Sep. 2016 – Jun. 2020

Research Areas and Directions

- **Artificial intelligence and intelligent control algorithms:** physics-constrained deep learning, reinforcement learning, adaptive dynamic programming, temporal neural networks, and digital twins.
- **Atomic spin inertial sensing and quantum precision measurement:** electron-spin-polarization stabilization and three-axis residual-magnetic-field compensation in atomic gyroscopes.
- **Multiphysics modeling and disturbance compensation:** atomic-ensemble dynamics modeling and state estimation, together with online inversion and coordinated compensation of coupled magnetic, optical, and thermal disturbances.

Academic Profile

- Research focuses on artificial intelligence, deep learning, and reinforcement-learning algorithms for atomic spin inertial sensing and quantum precision measurement, with particular emphasis on physics-informed deep learning, adaptive dynamic programming, temporal neural networks, and digital twins for complex-system modeling, state estimation, multiphysics disturbance compensation, and intelligent control.
- Authored 20 published or accepted papers in leading SCI-indexed journals, including *Science China Information Sciences* and *IEEE Transactions on Systems, Man, and Cybernetics: Systems*; 13 SCI papers are first- or corresponding-author works, including three papers in CAS Tier 1 Top journals and one ESI Highly Cited Paper; additionally published one first-author paper in *Fundamental Research*.
- Filed six Chinese invention patents; contributed to an atomic spin inertial-sensing prototype with performance at the internationally reported state of the art.

Research Projects

- **National Natural Science Foundation of China, Young Scientists Fund (Category C), Principal Investigator:** Physics-Constrained Deep-Learning Compensation of Multiphysics Coupled Disturbances in Atomic Spin Inertial Measurement; Grant No. 62603043, CNY 300,000.
- **2026 Hangzhou Soft Science Research Program (General Project), Principal Investigator:** Research on Innovation Mechanisms for AI Application Scenarios in Hangzhou.
- **STI 2030 Major Project "Quantum Communication and Quantum Computing" (Participant):** Ultrahigh-Sensitivity Atomic Spin Inertial Measurement Device and High-Precision Atomic Gyroscope Proof-of-Principle Prototype; prototype lead responsible for system identification, closed-loop electron-spin control under multiphysics disturbances, and performance targets.
- **National Key R&D Program of China, Participant:** High-Precision Atomic Spin Gyroscope Technology; prototype lead responsible for active magnetic-field compensation, high-precision vapor-cell temperature control, automated testing software, embedded electronics, and performance targets.

Research Publications

Fourteen representative first- or corresponding-author journal articles:

[1] **F. Li**, Z. Wang, F.-Y. Wang, et al. Digital Twin-Enabled Parallel Stabilization of Electron Spin Polarization in Optically Pumped Hybrid Spin Ensembles. *Science China Information Sciences*, 2026. **(Jointly sponsored by the Chinese Academy of Sciences and the National Natural Science Foundation of China; JCR Q1; IF 8.1; CAS Journal Ranking Tier 1, Top; CCF-A)**

[2] **F. Li**, Z. Wang, H. Pang, et al. Stochastic Spin Modeling and ANEU-TCN for Directional Magnetic Noise Estimation in Atomic Comagnetometers. *Fundamental Research*, 2026. **(Supervised and sponsored by the National Natural Science Foundation of China; JCR Q1; IF 7.0)**

[3] **F. Li**, J. Liu, Z. Wang, et al. Belief- and Risk-Driven Interval Type-2 Neuro-Fuzzy Output-Feedback Control for Partially Observed Stochastic Nonlinear Systems. *IEEE Transactions on Fuzzy Systems*, 2026. DOI: 10.1109/TFUZZ.2026.3745076. **(JCR Q1, IF 10.2; CAS Journal Ranking Tier 1, Top)**

[4] **F. Li**, Z. Wang, D. Liu, et al. Machine Learning for Quantum Sensing Systems: An Integrated Survey of Preparation, Measurement, Inference, and Control. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. DOI: 10.1109/TSMC.2026.3744356. **(JCR Q1, IF 8.7)**

[5] **F. Li**, Z. Wang, M. Zhang, Q. Zheng. Basis-Adaptive Spectral Optimization via Learnable Graph Fourier Bases. *Applied Soft Computing*, 2027, 204: 116607. **(JCR Q1, IF 7.8)**

[6] **F. Li**, Z. Wang, R. Wang, et al. Magnetic Field Compensation Control for Spin-Exchange Relaxation-Free Comagnetometer Using Reinforcement Learning. *IEEE Transactions on Instrumentation and Measurement*, 2023, 72: 9511010. **(JCR Q1, IF 7.0)**

[7] **F. Li**, H. Pang, Z. Wu, et al. Spin-Decoupling Calibration Method for Coil Constants in Optically Pumped Comagnetometers. *Measurement*, 2026, 257: 118819. **(JCR Q1, IF 6.1)**

[8] **F. Li**, Z. Wang, R. Wang, et al. Heating Temperature Control and Driving Voltage Nonlinear Feedback of LCVR for Stabilizing Laser Power in Atomic Spin Gyroscopes. *Nonlinear Dynamics*, 2025, 113(14): 18399–18411. **(JCR Q1, IF 5.7)**

[9] **F. Li**, Z. Wang, R. Wang, et al. Exponential Decay Adaptive Robust Control for Electron Spin Magnetization in Atomic Spin Gyroscopes. *Control Engineering Practice*, 2025, 163: 106405. **(JCR Q1, IF 5.3)**

[10] J. Liu, **F. Li** (corresponding author), Z. Wang, M. Zhang. Certified Worst-Case Fidelity Bounds for Quantum Gates under Structured Uncertainty. *Quantum Science and Technology*, 2026 (Accepted). **(JCR Q1, IF 4.9; CAS Journal Ranking Tier 2, Top)**

[11] **F. Li**, H. Pang, Z. Wang, et al. In Situ Real-Time Measurement for Electron Spin Polarization in Atomic Spin Gyroscopes. *iScience*, 2025, 28(2): 111757. **(JCR Q1, IF 4.5)**

[12] **F. Li**, M. Zhang, Z. Wang, J. Li, R. Wang. PI-FAS-CLO: Physics-Informed Fully Actuated System-Coordinate Learning Observer. *International Journal of Systems Science*, 2026 (Accepted). **(JCR Q1, IF 4.3)**

[13] **F. Li**, Z. Wang, W. Fan, et al. Effects of Stochastic Magnetic Noise on the SERF Co-Magnetometer. *Advanced Quantum Technologies*, 2024, 7(6): 2400040. **(JCR Q2, IF 4.3)**

[14] J. Li, Z. Wang, **F. Li** (corresponding author), et al. Physics-Informed Neural-Network Calibration of Pump–Probe Nonorthogonality in SERF Co-Magnetometers. *Measurement Science and Technology*, 2026, 37: 325902. **(JCR Q1, IF 3.7)**

Honors and Awards

Outstanding Graduate, Beihang University 2025
Outstanding Graduate Student, Beihang University 2024
Outstanding Graduate Academic Innovation Achievement Award, Beihang University 2024
First-Class Academic Scholarship, Beihang University, four consecutive years 2021–2024
Outstanding Research Team Award, Institute of Large-scale Scientific Facility, Beihang University, three consecutive years 2021–2023
First and Second Prizes, Shandong Division, National Undergraduate Electronic Design Contest 2018, 2019
First and Second Prizes, Shandong University Robot Competition 2017, 2018
Second Prize, Shandong Undergraduate Intelligent Control Competition 2017

Professional Service

- Editorial Board Member, *International Journal of Intelligent Control and Systems* (IJICS).
- Youth Editorial Board Member, *Advanced Information and Communication* (AIC).
- Youth Editorial Board Member, *Embodied and Biomimetic Intelligence* (EBI).
- Member, Chinese Association of Automation (CAA); Committee Member, CAA Technical Committee on Adaptive Dynamic Programming and Reinforcement Learning (TC-ADPRL).
- Track Co-chair, IEEE ISAICS 2026 (2026 IEEE 3rd International Symposium on AI and Cybersecurity).
- Reviewer for *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Instrumentation and Measurement*, and academic conferences including CCC and IACOP.